TED(21)-5033A **REVISION 2021**

Model Question Paper I

RENEWABLE ENERGY POWER PLANTS

Time: 3 Hour *Max.Marks: 75*

PART A

I. Answer all questions in one word or one sentence. Each question carries 1 mark.

(9*1=9 marks)

1	List any two non-conventional sources of energy	M1.01	R
2	Match the followings : <table><tr><td>(i) Pelton wheel</td><td>(a)Medium head</td></tr><tr><td>(ii) Francis turbine</td><td>(b)Low head</td></tr><tr><td>(iii) Kaplan turbine</td><td>(c)High Head</td></tr></table>	M1.02	R
3	Write any two advantages of solar energy	M2.01	R
4	Define solar irradiation ?	M2.01	R
5	Equation for power output from a wind turbine is_____________	M3.03	R
6	The speed at which power conversion stops in wind systems is called __________	M3.02	R
7	List out any two scheme for wind power generation	M3.04	R
8	Write any two disadvantages of tidal energy	M4.01	R
9	Enumerate types of fuel cells	M4.03	R

PART B

II. Answer any eight questions from the following, each question carries 3 marks.

(8*3=24 marks)

1	Write the classification of small hydel power plants based on capacity and water head	M1.02	U
2	Draw the structure of s solar cell	M2.02	R
3	Explain the concept of maximum power point tracking technique in PV system	M2.03	U
4	Draw the block diagram of a Grid connected solar PV system and label various blocks.	M2.03	R
5	Summarize the principle of wind energy conversion with a diagram	M 3.01	U
6	Draw the schematic diagram of a vertical axis wind energy conversion systems	M3.02	R
7	Determine the power in the wind if the wind speed is 20 m/s and blade length is 50 m .Take air density as 1.23 kg/m3	M3.03	A
8	Illustrate electrical power generation using tidal energy	M4.01	U
9	Explain the concept of Ocean Thermal Power Generation	M4.02	U
10	List any three applications of fuel cells	M4.03	R

PART C

Answer ALL questions. Each question carries 7 marks.

(6*7=42 marks)

III	With a neat diagram explain vapour dominated geothermal power plant.	M1.03	U
	OR		
IV	Distinguish between dome type and drum type biogas plants.	M1.04	U

V	Draw a neat sketch of a small hydro power plant and identify the functions of various parts.	M1.02	U
	OR		
VI	With a neat diagram explain flash steam geothermal power plant.	M1.03	U
VII	A house has the following electrical appliance usage ● One 18 watt fluorescent lamp used 5 hours per day. ● One 60 watt fan used for 6 hours per day. ● One 150 Watt Television 5 hours per day. Calculate (a) Daily solar power requirement (a) Ah rating of the 12 V Batteries required for 3 days of autonomy in a solar PV System. Assume 85% battery efficiency and 60% as depth of discharge.	M2.04	A
	OR		
VIII	Calculate(a) the rating of an inverter and (b) number of 350 Wp solar panels for a Solar PV System for the following load . Assume power generation factor of a solar panel = 3 and 30 % losses ● Ten 15 Watt CFL used 10 hours per day. ● Ten 60 Watt fan used for 12 hours per day. ● Eight 5w LED light used for 6 hours per day ● Two cell phone chargers 5w used for 5 hours per day	M2.04	A
IX	Illustrate the concept of solar pond.	M2.02	U
	OR		
X	Explain stand-alone solar PV systems with the help of a neat sketch.	M2.03	U
XI	State and explain any seven factors to be considered in selection of sites for wind power plant	M3.01	U
	OR		
XII	Classify and explain various types of wind mills	M3.02	U
XIII	Draw a neat sketch of oscillating water columns in tidal power generation.	M4.01	R

	OR		
XIV	Draw the block diagram of MHD power plant.	M4.04	R

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Model Question Paper II

RENEWABLE ENERGY POWER PLANTS

Time: 3 Hour *Max.Marks: 75*

PART A

I. Answer all questions in one word or one sentence. Each question carries 1 mark.
(9*1=9 marks)

1	Define Renewable energy?	M1.01	R
2	State any two advantages of small hydel power plants	M1.02	R
3	State the the purpose of a heliostat?	M2.02	R
4	List any two factors considered for installing a solar panel	M2.02	R
5	State any two applications of wind energy.	M3.01	R
6	Write any two classification of wind energy system.	M3.02	R
7	Wind speed at which wind turbine starts delivering shaft power is called __________.	M3.02	R
8	Difference in water level between high tide and low tide of an area is called _________	M4.01	R
9	The process or technology for producing energy by harnessing the temperature differences between ocean surface waters and deep ocean waters is called _________________________	M4.02	R

PART B

II. Answer any eight questions from the following, each question carries 3 marks.
(8*3=24 marks)

1	Outline the importance of renewable energy sources.	M1.01	U
2	List the criterias for selection of site for small hydro power plants	M1.02	R
3	Draw a neat sketch of a fixed dome type biogas plant.	M1.04	R
4	List the classification of geothermal resources	M1.03	R
5	Explain with a diagram operation of a solar power tower.	M2.02	U

6	Classify solar PV systems	M2.03	U
7	Compare Horizontal and Vertical axis windmills.	M3.02	U
8	Illustrate power generation from tide using single basin system	M4.01	U
9	Explain power generation from wave using buoys with a neat diagram	M4.01	U
10	Outline the concept of ocean thermal energy conversion	M 4.02	U

PART C

Answer ALL questions. Each question carries 7 marks.

(6*7=42 marks)

III	Draw the schematic diagram of binary cycle geothermal power plant and explain the operations.	M1.03	U
	OR		
IV	Explain the working of floating drum type biogas plant with a neat sketch.	M1.04	U
V	A house has the following electrical appliance usage: One 18 Watt fluorescent lamp with electronic ballast used 4 hours per day. One 60 Watt fan used for 2 hours per day. One 75 Watt refrigerator that runs 24 hours per day with compressor run 12 hours and off 12 hours. The system will be powered by 12Vdc, 110 Wp PV module. Determine the following: - 1.Power consumption demands 2.Size of the PV panel 3.Rating of Inverter (Assume panel generation factor=3.4 and a 30 % loss)	M2.04	A
	OR		
VI	4 Nos of 12 V, 100 Ah batteries connected in parallel are to be charged from a solar panel. Calculate the Number of 110 Wp solar panels required for this purpose .Assume charging current = 1/10 of ampere hour rating	M2.04	A

VII	Explain grid connected photovoltaic system with a neat block diagram.	M2.03	U
	OR		
VIII	With a neat sketch explain the concept of solar pond.	M2.02	U
IX	Explain with a neat sketch the working of a horizontal axis wind mill	M3.02	U
	OR		
X	Explain stand-alone wind energy systems with block diagram	M3.03	U
XI	Explain the Constant Speed Constant Frequency Scheme for wind power generation.	M3.04	U
	OR		
XII	With the help of a block diagram explain basic components of wind energy conversion system	M3.02	U
	OR		
XIII	Explain the principle of operation MHD power generation with a neat sketch.	M4.04	U
	OR		
XIV	Summarize the operation of closed cycle ocean thermal energy conversion with a block diagram	M4.02	U